

1. Independence

E and S are independent if $P(E \text{ and } S) = P(E) P(S)$.

When $P(S) > 0$, this is equivalent to $P(E | S) = P(E)$.

Suppose 50 of all 100 students take economics.

$P(E) = 0.50$, while $P(E | S) = 0.60$: not independent.

2. Read the two-way table

	Economics	Not economics	Total
Statistics	24	16	40
Not statistics	26	34	60
Total	50	50	100

$24 / 40 = 0.60$ differs from $50 / 100 = 0.50$.